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Roll No

EX-505 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Power System - I
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. The yearly load duration curve of a power plant is a straight line. The maximum load is 500MW and the minimum load is 400MW. The capacity of the plant is 750MW. Find the
 - i) Plant Capacity factor
 - ii) Load factor
 - iii) Utilization factor
 - iv) Reserve capacity
2. A power station is to supply for regions of load whose peak loads are 10 MW, 5 MW, 8 MW and 7 MW. The diversity factor of the load at the station is 1.5 and the average load factor is 0.6 Calculate:
 - i) Maximum demand on the station.
 - ii) Annual energy supplied from the station.Suggest the installed capacity and the number of units taking all aspects into account.
3. What is the percentage saving in feeder copper if the line voltage in a 2 wire d.c. system be raised from 220V to 400V for the same power transmitted over same distance and having the same power loss.
4. Explain the following:
 - i) Classification of cables
 - ii) Dielectric stress in cable
 - iii) Grading of cables
5. State and prove Kelvin's law for most economical size of conductor for transmission. Discuss its limitations.
6. A three phase 132kV transmission line is connected to a 50mW load at a power factor of 0.85 lagging. The line constants of the 80km long line are $Z = 96 \angle 78^\circ \Omega$ and $Y = 0.0D1 \angle 90^\circ S$. Using nominal T-circuit representation calculate A, B, C and D constants of the line.

OR

How are the transmission line insulators classified? What are the causes of failure of insulators.

7.
 - a) Explain 3-phase, 4-wire system of distributing electrical power.
 - b) What is a substation? How are sub-stations classified? Draw lay-out of a sub-station.
8. Write short notes on any two:
 - a) Ferranti effect
 - b) Composite conductors
 - c) Corona effect
 - d) Feeder

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